Data types and notation

BME423 · V6. Population: the entire group; sample: the measured subset. A population parameter is fixed; a sample statistic changes across samples.

x_i (observation i); n (sample size); $\bar{x}$ (sample mean); μ (population mean); s (sample SD); σ (population SD); s^2, σ^2 (variances). $\sum$ means add the specified terms.

Interval–continuous: measurements on a continuum, e.g. blood pressure, height or membrane potential. **Interval–discrete:** counts or rounded measurements, e.g. tender joints, text-message count or age rounded to years. **Categorical–nominal:** unordered categories, e.g. country or ZIP code. **Categorical–ordered:** categories with a natural order, e.g. cancer stage or severity. A numeric label such as ZIP code is not an interval measurement.

Height data: mean, variance, SD and SEM

Question. Heights are 65, 69, 59, 64, 62, 67, 68, 62, 64, 65. Estimate the mean, variance, SD and standard error of the mean.

1. Add the observations; divide by their count

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$n = 10, \quad \sum x_i = 645$$

$$\bar{x} = \frac{645}{10} = 64.5$$

2. Subtract the mean, then square each difference

$$d_i = x_i - \bar{x}, \qquad d_i^2 = (x_i - \bar{x})^2$$

$$d_1 = 65 - 64.5 = .5, \quad d_1^2 = .5^2 = .25$$

$$d_2 = 69 - 64.5 = 4.5, \quad d_2^2 = 4.5^2 = 20.25$$

$$d_3 = 59 - 64.5 = -5.5, \quad d_3^2 = (-5.5)^2 = 30.25$$

$$d_4 = 64 - 64.5 = -.5, \quad d_4^2 = (-.5)^2 = .25$$

$$d_5 = 62 - 64.5 = -2.5, \quad d_5^2 = (-2.5)^2 = 6.25$$

$$d_6 = 67 - 64.5 = 2.5, \quad d_6^2 = 2.5^2 = 6.25$$

$$d_7 = 68 - 64.5 = 3.5, \quad d_7^2 = 3.5^2 = 12.25$$

$$d_8 = 62 - 64.5 = -2.5, \quad d_8^2 = (-2.5)^2 = 6.25$$

$$d_9 = 64 - 64.5 = -.5, \quad d_9^2 = (-.5)^2 = .25$$

$$d_{10} = 65 - 64.5 = .5, \quad d_{10}^2 = .5^2 = .25$$

$$\sum d_i^2 = 82.5$$

3. Divide by one less than the sample size

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{82.5}{10 - 1} = 9.16666667$$

4. Square-root variance; scale SD for a mean

$$s = \sqrt{s^2}, \qquad SEM = \frac{s}{\sqrt{n}}$$

$$s = \sqrt{9.16666667} = 3.02765035$$

$$SEM = \frac{3.02765035}{\sqrt{10}} = .957427108$$

SD describes variation among heights; SEM describes variation among repeated sample means. Variance has squared units; SD and SEM have the original units.

Whole population instead: use its population mean and divide squared deviations by N, not $n - 1$.

$$\mu = \frac{\sum_i x_i}{N}, \qquad \sigma^2 = \frac{\sum_i (x_i - \mu)^2}{N}$$

Sorted heights: median, quartiles and IQR

Question. For the same heights, find the median, quartiles, IQR and range. Sorted data: 59, 62, 62, 64, 64, 65, 65, 67, 68, 69.

1. Locate a percentile between sorted ranks

$$R = 1 + \frac{p(n-1)}{100} = I + d$$

$$R_{25} = 1 + \frac{25(10-1)}{100} = 3.25$$

$$R_{50} = 1 + \frac{50(10-1)}{100} = 5.5$$

$$R_{75} = 1 + \frac{75(10-1)}{100} = 7.75$$

p (percentile number); I (integer rank); d (fraction of the gap to the next rank).

2. Interpolate the measured values, not the ranks

$$x_p = x_{(I)} + d[x_{(I+1)} - x_{(I)}]$$

$$Q_1 = 62 + .25(64 - 62) = 62.5$$

$$Q_2 = 64 + .5(65 - 64) = 64.5$$

$$Q_3 = 65 + .75(67 - 65) = 66.5$$

3. Subtract the appropriate endpoints

$$IQR = Q_3 - Q_1, \quad \text{range} = x_{\max} - x_{\min}$$

$$IQR = 66.5 - 62.5 = 4$$

$$\text{range} = 69 - 59 = 10$$

Discrete table: probabilities, mean and variance

Question. Check this probability function. Find its mean, variance, CDF, and the probability that the outcome is at least 2 but less than 7.

x	-1	2	3	7
$p(x)$	.55	.15	.10	.20

1. Check both probability requirements

$$0 \le p(x) \le 1, \qquad \sum_x p(x) = 1$$

$$.55 + .15 + .10 + .20 = 1$$

2. Weight each value by its probability

$$\mu = E[X] = \sum_x x\,p(x)$$

$$\mu = (-1)(.55) + 2(.15) + 3(.10) + 7(.20) = -.55 + .30 + .30 + 1.40 = 1.45$$

X (random variable before observing the outcome); x (one possible value); $p(x)$ (its probability).

3. Square the values before weighting

$$E[X^2] = \sum_x x^2 p(x)$$

$$E[X^2] = 1(.55) + 4(.15) + 9(.10) + 49(.20) = .55 + .60 + .90 + 9.80 = 11.85$$

4. Direct variance: weight EVERY squared deviation

$$v_x = (x - \mu)^2 p(x), \quad \text{Var}(X) = \sum_x v_x$$

$$v_{-1} = (-1 - 1.45)^2(.55) = 3.301375$$

$$v_2 = (2 - 1.45)^2(.15) = .045375$$

$$v_3 = (3 - 1.45)^2(.10) = .24025$$

$$v_7 = (7 - 1.45)^2(.20) = 6.1605$$

$$\text{Var}(X) = 3.301375 + .045375 + .24025 + 6.1605 = 9.7475$$

5. Check with the second-moment shortcut

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = 11.85 - 1.45^2 = 9.7475$$

$$SD(X) = \sqrt{9.7475} = 3.12209865$$

6. Add only the included masses

$$P(X \in A) = \sum_{x \in A} p(x)$$

$$P(2 \le X < 7) = p(2) + p(3) = .15 + .10 = .25$$

Discrete CDF and graph endpoints

1. Accumulate probabilities through the endpoint

$$F(x) = P(X \le x) = \sum_{t \le x} p(t)$$

$$F(x) = \begin{cases} 0 & x < -1 \\ .55 & -1 \le x < 2 \\ .70 & 2 \le x < 3 \\ .80 & 3 \le x < 7 \\ 1 & x \ge 7 \end{cases}$$

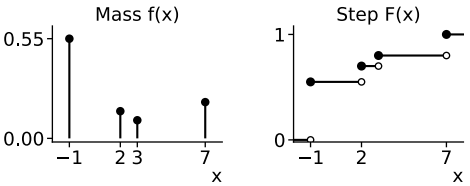
2. Include or exclude the endpoint mass

$$P(X > a) = 1 - F(a)$$

$$P(X > 2) = 1 - .70 = .30$$

$$P(X \ge 2) = 1 - P(X < 2) = 1 - .55 = .45$$

$$P(-1 < X < 3) = p(2) = .15$$



Sketch: PMF uses separate mass spikes. CDF jumps by each mass, stays flat between values, and includes the new height at the jump (filled dot); the lower endpoint is open. A discrete point can have positive probability.

Loaded die and repeated draws

Question. A die has $p(1) = .20$; each of 2–6 has probability .16. Find $P(X \le 3)$, mean and variance.

Weight using the stated masses

$$P(X \le a) = \sum_{x \le a} p(x), \quad E[X] = \sum x p(x)$$

$$P(X \le 3) = .20 + .16 + .16 = .52$$

$$E[X] = 1(.20) + (2 + 3 + 4 + 5 + 6)(.16) = 3.4$$

$$E[X^2] = 1(.20) + (4 + 9 + 16 + 25 + 36)(.16) = 14.6$$

$$\text{Var}(X) = E[X^2] - \mu^2 = 14.6 - 3.4^2 = 3.04$$

Continuous density: integrate, then evaluate bounds

Question. Let $f(x) = x/2$ for $0 < x < 2$, zero elsewhere. Verify the density, then find its CDF, mean, variance, a tail probability and median.

1. Check total area and nonnegative height

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^2 \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_0^2 = \frac{2^2}{4} - \frac{0^2}{4} = 1$$

The density is nonnegative on its support. Density height is not probability; probability is area. $\int_a^b g(x)\,dx = G(b) - G(a)$ means evaluate the antiderivative at the upper bound, then subtract its lower-bound value.

2. Accumulate area only up to the requested value

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$F(x) = \int_0^x \frac{t}{2}\,dt = \left[\frac{t^2}{4}\right]_0^x = \frac{x^2}{4}$$

$$F(x) = \begin{cases} 0 & x \le 0 \\ x^2/4 & 0 < x < 2 \\ 1 & x \ge 2 \end{cases}$$

3. Multiply by the outcome before integrating

$$\mu = E[X] = \int x f(x)\,dx$$

$$E[X] = \int_0^2 x\frac{x}{2}\,dx = \int_0^2 \frac{x^2}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} - 0 = \frac{4}{3}$$

4. Use a second moment for variance

$$E[X^2] = \int x^2 f(x)\,dx, \quad \sigma^2 = E[X^2] - \mu^2$$

$$E[X^2] = \int_0^2 \frac{x^3}{2}\,dx = \left[\frac{x^4}{8}\right]_0^2 = 2$$

$$\sigma^2 = 2 - \left(\frac{4}{3}\right)^2 = \frac{18 - 16}{9} = \frac{2}{9}$$

5. A right tail is the unaccumulated area

$$P(X > a) = 1 - F(a)$$

$$F(1.5) = \frac{1.5^2}{4} = .5625$$

$$P(X > 1.5) = 1 - .5625 = .4375$$

6. A median splits the total area in half

$$F(m) = .5$$

$$\frac{m^2}{4} = .5 \Rightarrow m^2 = 2 \Rightarrow m = \sqrt{2}$$

Keep the positive root inside the support. For a continuous variable, $P(X = a) = 0$; strict and inclusive endpoints have the same probability.

A density with two pieces

Question. A triangular density is $f(x) = x$ on (0,1) and $f(x) = 2 - x$ on [1,2). Find its CDF and $P(.5 < X < 1.5)$.

Carry the completed first area into the second piece

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$0 < x < 1: \quad F(x) = \int_0^x t\,dt = \frac{x^2}{2}$$

$$1 \le x < 2: \quad F(x) = \frac{1}{2} + \int_1^x (2 - t)\,dt$$

$$= \frac{1}{2} + \left[2t - \frac{t^2}{2}\right]_1^x = 2x - \frac{x^2}{2} - 1$$

$$F(x) = 0\ (x \le 0), \qquad F(x) = 1\ (x \ge 2)$$

Subtract accumulated areas

$$P(a < X < b) = F(b) - F(a)$$

$$F(.5) = \frac{.5^2}{2} = .125$$

$$F(1.5) = 2(1.5) - \frac{1.5^2}{2} - 1 = .875$$

$$P(.5 < X < 1.5) = .875 - .125 = .75$$

The CDF remains continuous at 1; only its formula changes. Do not restart accumulated probability at zero.

Successive events: update the available outcomes

Question. Two cards are drawn without replacement from 52 cards. Find the probability that both are kings.

Without replacement: update the second draw

$$P(A \cap B) = P(A)P(B \mid A)$$

$$P(\text{two kings}) = \frac{4}{52}\frac{3}{51} = .00452489$$

For independent events, the second probability stays unchanged. Three fair coin flips have 8 equally likely sequences; exactly two heads occurs as HHT, HTH or THH, so its probability is 3/8.

Homework density: solve the unknown endpoint

Question. $f(x) = .1 + x$ for $0 < x \le c$, zero elsewhere. Find c, the CDF, mean, variance and requested probabilities.

1. Force the full support area to equal one

$$\int_0^c f(x)\,dx = 1$$

$$\int_0^c (.1 + x)\,dx = \left[.1x + \frac{x^2}{2}\right]_0^c = 1$$

$$.1c + \frac{c^2}{2} = 1 \Rightarrow c^2 + .2c - 2 = 0$$

2. Solve the quadratic; keep the valid endpoint

$$c = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$c = \frac{-.2 \pm \sqrt{.2^2 - 4(1)(-2)}}{2}$$

$$= \frac{-.2 \pm \sqrt{8.04}}{2}$$

$$c = 1.31774469$$

Here the quadratic coefficients are $a = 1, b = .2, c_{\text{quad}} = -2$; the support endpoint is the positive root. A negative endpoint cannot give this support.

3. Integrate only to the requested value

$$F(x) = \int_0^x f(t)\,dt$$

$$F(x) = \left[.1t + \frac{t^2}{2}\right]_0^x = .1x + \frac{x^2}{2}$$

$$F(x) = 0\ (x \le 0), \qquad F(x) = 1\ (x \ge c)$$

4. Multiply by x and expand before integrating

$$E[X] = \int_0^c x f(x)\,dx$$

$$x(.1 + x) = .1x + x^2$$

$$E[X] = \int_0^c (.1x + x^2)\,dx$$

$$= \left[.05x^2 + \frac{x^3}{3}\right]_0^c$$

$$= .05(1.31774469)^2 + \frac{(1.31774469)^3}{3}$$

$$= .849555608$$

5. Square x, not the entire density

$$E[X^2] = \int_0^c x^2 f(x)\,dx$$

$$x^2(.1 + x) = .1x^2 + x^3$$

$$E[X^2] = \left[\frac{x^3}{30} + \frac{x^4}{4}\right]_0^c$$

$$= \frac{(1.31774469)^3}{30} + \frac{(1.31774469)^4}{4}$$

$$= .830088878$$

6. Subtract squared mean from second moment

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = .830088878 - (.849555608)^2$$

$$= .108344148$$

7. Use the CDF and complement

$$P(X > a) = 1 - F(a), \quad P(X < b) = F(b)$$

$$P(X > .5) = 1 - [.1(.5) + .5^2/2] = .825$$

$$P(X < .6) = .1(.6) + .6^2/2 = .24$$

8. Transform the mean without a new density

$$E[aX + b] = aE[X] + b$$

$$E[2X + 1] = 2(.849555608) + 1 = 2.69911122$$

9. Find the median using half the total area

$$F(m) = .5$$

$$.1m + \frac{m^2}{2} = .5 \Rightarrow m^2 + .2m - 1 = 0$$

$$m = \frac{-.2 + \sqrt{.04 + 4}}{2} = .904987562$$

Keep the root inside the support, 0 to 1.31774469.

Uniform range and unknown coefficient

Question. For the standard uniform distribution on (0,1), find its height, the probability below .5, its mean and variance.

Spread total area one across the interval

$$f(x) = \frac{1}{b-a}, \quad P(u < X < v) = \frac{v-u}{b-a}$$

$$f(x) = \frac{1}{1-0} = 1, \qquad P(0 < X < .5) = \frac{.5-0}{1-0} = .5$$

$$E[X] = \frac{a+b}{2} = \frac{0+1}{2} = .5$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{(1-0)^2}{12} = \frac{1}{12}$$

Unknown coefficient: factor it out of the area

$$f(x) = kx\ (0 < x < 2), \quad k\int_0^2 x\,dx = 1$$

$$k\left[\frac{x^2}{2}\right]_0^2 = 1 \Rightarrow 2k = 1 \Rightarrow k = \frac{1}{2}$$

Glucose: left, between and right Normal areas

Question. Glucose $X \sim N(95, 8.7^2)$. Find the probabilities below 85, from 85 to 115, and above 115. Here 8.7 is SD, not variance.

Write the Normal density and accumulated area

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$f_X(x) = \frac{1}{8.7\sqrt{2\pi}} \exp\left[-\frac{(x-95)^2}{2(8.7)^2}\right]$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

The CDF integral need not be solved algebraically; use the Normal table after standardizing.

1. Convert measurement cutoffs to SD units

$$z = \frac{x - \mu}{\sigma}$$

$$z_{85} = \frac{85 - 95}{8.7} = -1.14942529 \approx -1.15$$

$$z_{115} = \frac{115 - 95}{8.7} = 2.29885057 \approx 2.30$$

2. Read accumulated LEFT area in the Normal table

$$\Phi(z) = P(Z \le z), \quad \Phi(-z) = 1 - \Phi(z)$$

$$\Phi(1.15) = .8749 \Rightarrow \Phi(-1.15) = .1251$$

$$\Phi(2.30) = .9893$$

For 1.15 use row 1.1 and column .05; for 2.30 use row 2.3 and column .00. Shade the requested side before subtracting.

3. Use the CDF for each requested region

$$P(X < a) = \Phi(z_a)$$

$$P(X < 85) = \Phi(-1.15) = .1251$$

$$P(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

$$P(85 < X < 115) = .9893 - .1251 = .8642$$

$$P(X > b) = 1 - \Phi(z_b)$$

$$P(X > 115) = 1 - .9893 = .0107$$

TI-84: DISTR → normalcdf(lower,upper,95,8.7). Use −1E99/1E99 for infinite bounds. Calculator values are .1251903, .8640530 and .0107567; small table-rounding differences are expected.

$$z_0 = \frac{0 - 95}{8.7} = -10.9195402$$

$$P(X < 0) = \Phi(-10.9195402) \approx 0, \quad P(X = 95) = 0$$

A single exact value has zero area. The three disjoint regions below 85, from 85 to 115, and above 115 sum to one.

Given a Normal area: recover the measurement

Question. Find the glucose 25th and 75th percentiles and central 95% interval.

1. Match cumulative area to a standard cutoff

$$\Phi(z_p) = p, \qquad x_p = \mu + z_p\sigma$$

$$z_{.25} = -.67448975, \quad z_{.75} = +.67448975$$

$$x_{25} = 95 + (-.67448975)(8.7) = 89.1319392$$

$$x_{75} = 95 + (.67448975)(8.7) = 100.868061$$

2. Put half the omitted area in each tail

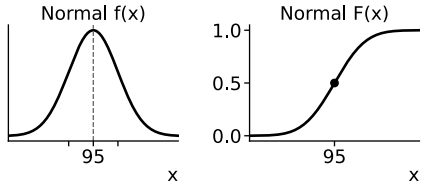
$$P(-z_* < Z < z_*) = .95 \Rightarrow \Phi(z_*) = .975$$

$$z_* = 1.96$$

$$x_L = \mu - z_*\sigma = 95 - 1.96(8.7) = 77.948$$

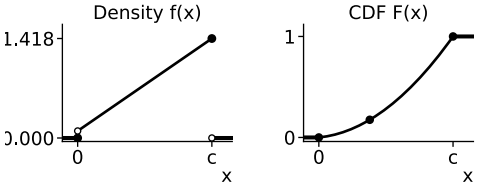
$$x_U = x_{97.5} = \mu + z_*\sigma = 95 + 1.96(8.7) = 112.052$$

TI-84: invNorm(left area,95,8.7). A right-tail area .025 means LEFT area .975. Use the specified tail; do not enter a percentage such as 97.5.



Sketch: density is a bell centered at μ; mark cutoffs and shade area. Normal CDF rises smoothly from 0 to 1, crosses .5 at μ, and has no jumps. Larger SD widens the density.

Draw a continuous PDF and CDF



Density $f(x) = .1 + x$**:** zero outside (0,*c*]; inside, draw a straight rising line from height .1 to .1+*c*. Shade an interval under the line. $f(.5) = .6$ is a height, not a probability.

CDF: zero before 0, curve $F(x) = .1x + x^2/2$ inside, then flat at 1 after *c*. It is continuous and increasing. Its slope at each point is the density height. Changing formulas can create corners; a continuous density has no point-mass jumps.

Average of n: sampling distribution and CLT

Question. Individual responses have mean 4 and SD 1.5. For independent samples of 50, find the probability that the sample mean is between 3.5 and 3.8.

1. Keep the center; reduce variance for averaging

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$E[\bar{X}] = 4, \quad \text{Var}(\bar{X}) = \frac{1.5^2}{50} = .045$$

$$SD(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = .212132034$$

$\bar{X}$ (random sample mean); $\bar{x}$ (one observed mean). If individuals are Normal, means are Normal at any sample size. Otherwise the CLT gives an approximately Normal mean for sufficiently large independent samples (course rule: $n \ge 30$).

2. Divide each mean cutoff by the MEAN spread

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$z_L = \frac{3.5 - 4}{.212132034} = -2.35702260$$

$$z_U = \frac{3.8 - 4}{.212132034} = -.942809042$$

3. Subtract the accumulated areas

$$P(a < \bar{X} < b) = \Phi(z_b) - \Phi(z_a)$$

$$\Phi(-.942809042) = .172889293$$

$$\Phi(-2.35702260) = .009211063$$

$$P(3.5 < \bar{X} < 3.8) = .172889293 - .009211063$$

$$= .163678230$$

Rounded table cutoffs −.94 and −2.36 give .1736−.0091=.1645. For an individual observation, use σ itself; do not divide it by $\sqrt{n}$.

Shift, scale, and independent sums

Question. For the homework glucose variable, find $E[X - 26]$. A constant changes the center, while multiplying a variable also changes its spread.

1. Apply coefficients directly to the mean

$$E[aX + b] = a\mu + b, \quad \text{Var}(aX + b) = a^2\sigma^2$$

$$E[X - 26] = 95 - 26 = 69$$

$$\text{Var}(X - 26) = 1^2(8.7^2) = 75.69$$

2. For independent variables, ADD variances

$$E[aX + bY] = a\mu_X + b\mu_Y$$

$$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2$$

For two independent measurements with the glucose distribution:

$$E[X + Y] = 95 + 95 = 190$$

$$\text{Var}(X + Y) = 8.7^2 + 8.7^2 = 151.38$$

$$\text{Var}(X - Y) = 8.7^2 + (-1)^2(8.7^2) = 151.38$$

Subtraction still adds variance because its coefficient is squared. The independent-sum variance rule does not apply to $X + X$: that is $2X$, whose variance is $4\sigma^2$.

Sample size and precision of an average

Question. The height sample has SD 3.02765035 and SEM .957427108 with 10 observations. How does the estimated SEM change with four times as many observations?

Compare sample sizes with individual spread held fixed

$$SEM_{\text{new}} = SEM_{\text{old}}\sqrt{\frac{n_{\text{old}}}{n_{\text{new}}}}$$

$$n_{\text{new}} = 4(10) = 40$$

$$SEM_{\text{new}} = .957427108\sqrt{\frac{10}{40}} = .478713554$$

Quadrupling sample size halves SEM. For a target SEM, rearrange $SEM = s/\sqrt{n}$: $n = (s/SEM_{\text{target}})^2$, then round upward. Larger samples improve the mean's precision; they do not necessarily reduce individual variation.

One binary proportion: SD versus SE

Question. Eight of 61 patients have the event. Estimate the event proportion and its SE.

Use event count divided by all patients

$$\hat{p} = \frac{x}{n}, \quad s_{\text{binary}} = \sqrt{\hat{p}(1-\hat{p})}$$

$$\hat{p} = \frac{8}{61} = .131147541$$

$$s_{\text{binary}} = \sqrt{.131147541(.868852459)} = .337561644$$

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$SE(\hat{p}) = \sqrt{\frac{.131147541(.868852459)}{61}} = .0432203397$$

The SE is uncertainty in the estimated proportion, not the spread of individual zero/one outcomes.

One mean: student LDL against a fixed reference

Question. Sixteen USC undergraduates have mean LDL 103 mg/dL and SD 16 mg/dL. The faculty/staff reference mean is 115 mg/dL. Assuming Normal independent responses, is the undergraduate population mean different at $\alpha = .05$?

1. State the population hypotheses

$$H_0 : \mu = \mu_0, \qquad H_A : \mu \neq \mu_0$$
$$H_0 : \mu = 115, \qquad H_A : \mu \neq 115$$

In words: the undergraduate population mean equals 115 under the null; it differs from 115 under the alternative. μ_0 is the fixed reference, not another sample mean.

2. Estimate the mean's standard error

$$SE = \frac{s}{\sqrt{n}}$$
$$SE = \frac{16}{\sqrt{16}} = \frac{16}{4} = 4$$

3. Standardize the observed departure

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{103 - 115}{4} = \frac{-12}{4} = -3$$

4. Use the one-sample degrees of freedom

$$\nu = n - 1, \qquad |t| > t_{\text{crit}} \Rightarrow \text{reject } H_0$$
$$\nu = 16 - 1 = 15$$
$$t_{\text{crit}} = t_{.025,15} = 2.131450$$
$$|t| = 3 > 2.131450$$

Table: row 15, total two-tail column .05 (printed cutoff 2.131). Reject the null: undergraduate population mean LDL differs from 115 at .05; the sample direction is lower.

p-value: $p = 2P(T_{15} \geq 3) = .00897274$. Compare p directly with .05. It is not the probability that the null hypothesis is true.

Two independent means: cognitive response

Question. Control: $n_1 = 22$, mean 49, SD 8.4. Intervention: $n_2 = 27$, mean 57, SD 9.5. Test whether the population means differ at .05. Assume independent Normal populations with a common variance.

1. State the hypotheses for the two populations

$$H_0 : \mu_1 = \mu_2, \qquad H_A : \mu_1 \neq \mu_2$$

In words: control and intervention have equal population means under the null; their population means differ under the alternative.

2. Pool VARIANCES, weighted by their df

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$s_1^2 = 8.4^2 = 70.56, \qquad s_2^2 = 9.5^2 = 90.25$$
$$s_p^2 = \frac{21(70.56) + 26(90.25)}{22 + 27 - 2}$$
$$= \frac{1481.76 + 2346.50}{47} = 81.4523404$$

3. Compute the standard error of the difference

$$SE_{\text{diff}} = \sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$
$$SE_{\text{diff}} = \sqrt{81.4523404\left(\frac{1}{22} + \frac{1}{27}\right)}$$
$$= 2.59212894$$

4. Divide the observed mean gap by its SE

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE_{\text{diff}}}$$
$$t = \frac{49 - 57}{2.59212894} = -3.08626622$$

5. Compare with a two-sided cutoff

$$\nu = n_1 + n_2 - 2, \quad |t| > t_{\text{crit}}$$
$$\nu = 22 + 27 - 2 = 47$$
$$|t| = 3.08626622 > t_{.025,47} = 2.01174051$$

If row 47 is missing, neighboring rows 46 and 48 give 2.013 and 2.011; the decision is unchanged. Reject equal population means. The intervention has the larger sample mean. $p = .00339345$. Unequal sample sizes do not prevent pooling when common variance is assumed.

Read supplied t-test output

```
Welch Two Sample t-test
t = -4.5298, df = 15.611
p-value = 0.0003625
mean of x   mean of y
64.5        69.7
```

Read t (observed statistic), df (reference distribution) and p-value. $.0003625 < .05$: reject equal population means. The negative statistic matches the first mean being smaller. Welch output can have fractional df; use the df printed in that output.

```
Two Sample t-test
t = -4.5298, df = 18
p-value = 0.0002594
```

"Two Sample t-test" with pooled variances uses $\nu = n_1 + n_2 - 2$. The course's hand calculations use the stated common-variance model; interpret supplied output according to its printed test name.

Three filgrastim doses: balanced ANOVA

Question. Fifteen patients per dose receive low, medium or high filgrastim. Their neutrophil concentration summaries are below. Test whether the three population means are all equal at .05; then compare dose pairs as one family.

Dose	Mean	SD
Low	3.7	1.1
Medium	4.6	1.4
High	5.2	2.1

1. State the overall population hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \text{not all population means are equal}$$

In words: the three dose population means are equal under the null; at least one differs under the alternative. Assume independent Normal responses and equal population variances.

2. Average the group variances for within spread

$$s_{\text{wit}}^2 = \frac{\sum_{j=1}^{m} s_j^2}{m}$$
$$s_{\text{wit}}^2 = \frac{1.1^2 + 1.4^2 + 2.1^2}{3}$$
$$= \frac{1.21 + 1.96 + 4.41}{3} = 2.52666667$$

$m = 3$ (number of groups); $n = 15$ (people per group); s_j (SD in group j).

3. Average the means; square their deviations

$$\bar{\bar{x}} = \frac{\sum_j \bar{x}_j}{m}$$
$$\bar{\bar{x}} = \frac{3.7 + 4.6 + 5.2}{3} = 4.5$$
$$\sum_j (\bar{x}_j - \bar{\bar{x}})^2 = (3.7 - 4.5)^2$$
$$+ (4.6 - 4.5)^2 + (5.2 - 4.5)^2$$
$$= .64 + .01 + .49 = 1.14$$

4. Divide by m−1, then multiply by group size

$$s_{\bar{x}}^2 = \frac{\sum_j (\bar{x}_j - \bar{\bar{x}})^2}{m - 1}, \quad s_{\text{bet}}^2 = n s_{\bar{x}}^2$$
$$s_{\bar{x}}^2 = \frac{1.14}{3 - 1} = .57$$
$$s_{\text{bet}}^2 = 15(.57) = 8.55$$

5. Divide between by within

$$F = \frac{s_{\text{bet}}^2}{s_{\text{wit}}^2}$$
$$F = \frac{8.55}{2.52666667} = 3.38390501$$

6. Use numerator and denominator df in that order

$$\nu_1 = m - 1, \quad \nu_2 = m(n - 1)$$
$$\nu_1 = 3 - 1 = 2, \quad \nu_2 = 3(15 - 1) = 42$$
$$F = 3.38390501 > F_{.05;2,42} = 3.21994229$$

Table: right-tail .05; numerator df 2 across the top, denominator df 42 down the side, lightface value 3.22. Reject all-equal population means; at least one dose population differs. $p = .04339471$. The pairwise calculations are separate from the overall decision.

Use supplied ANOVA output for follow-up tests

```
            Df Mean Sq
Group        2  389.7
Residuals   19   66.8
F value = 5.832
Pr(>F) = 0.0106
```

The residual Mean Sq is the within variance, 66.8; residual df is 19. The group Mean Sq is the between variance. $F = s_{\text{bet}}^2 / s_{\text{wit}}^2$; the reported F uses the unrounded mean squares. $.0106 < .05$: reject all-equal population means. For requested pairwise tests, retain within variance 66.8 and df 19; do not substitute a new two-group df.

Conditions: quantitative outcomes, independent observations, Normal populations and a common variance. The hand ANOVA above requires equal group sizes. A significant overall test establishes a difference somewhere; the adjusted comparisons identify which pairs are supported.

Filgrastim pairs: Bonferroni, every comparison

Question. Compare all three dose pairs at family $\alpha = .05$, using the original ANOVA within variance and residual df.

1. State the hypothesis for EACH dose pair

$$H_{0,ij} : \mu_i = \mu_j, \quad H_{A,ij} : \mu_i \neq \mu_j$$

In words: each null says the two named dose populations have equal means; each alternative says those population means differ.

2. Divide family alpha by the requested pair count

$$k = \frac{m(m-1)}{2}, \quad \alpha_{\text{pair}} = \frac{\alpha_{\text{family}}}{k}$$
$$k = \frac{3(3-1)}{2} = 3, \quad \alpha_{\text{pair}} = \frac{.05}{3} = .01666667$$
$$\nu = m(n-1) = 3(14) = 42$$
$$t_{\text{crit}} = t_{\alpha_{\text{pair}}/2,42} = 2.49366738$$

Use total two-tail probability .01666667, or one-tail .00833333, once. k counts the specified comparison family, not automatically every possible pair.

TI-84, missing table column: DISTR → invT. Enter the LEFT cumulative area and original residual df:

$$t_{\text{crit}} = \text{invT}(1 - \alpha_{\text{pair}}/2, \nu)$$
$$= \text{invT}(1 - (.05/3)/2, 42) = 2.49366738$$

3. Compute the shared SE using the original within variance

$$SE_{ij} = \sqrt{s_{\text{wit}}^2\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$
$$SE_{ij} = \sqrt{2.52666667\left(\frac{1}{15} + \frac{1}{15}\right)}$$
$$= \sqrt{\frac{2(2.52666667)}{15}} = .580421303$$

4. Low versus Medium

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{LM}| = \frac{|3.7 - 4.6|}{.580421303} = 1.550598$$
$$1.550598 < 2.49366738 \Rightarrow \text{do not reject } H_{0,LM}$$

5. Low versus High

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{LH}| = \frac{|3.7 - 5.2|}{.580421303} = 2.584330$$
$$2.584330 > 2.49366738 \Rightarrow \text{reject } H_{0,LH}$$

6. Medium versus High

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{MH}| = \frac{|4.6 - 5.2|}{.580421303} = 1.033732$$
$$1.033732 < 2.49366738 \Rightarrow \text{do not reject } H_{0,MH}$$

Conclusion: only low versus high is supported as different at family .05. A nonsignificant pair does not establish that its population means are identical.

Holm: compare treatments only with control

Question. The homework mortality table has three treatments and one control. The three Yates-corrected treatment–control statistics are 18.234103, 9.494427 and 4.689514. Use Holm at family .05.

1. Order the evidence, then reduce the remaining family

$$\alpha_j = \frac{\alpha}{k - j + 1}$$
$$k = 3: \quad \alpha_1 = .05/3 = .01666667$$
$$\alpha_2 = .05/2 = .025, \qquad \alpha_3 = .05/1 = .05$$

Order p-values from smallest to largest. For the same df, order χ^2 statistics largest to smallest. **In words:** each null says that treatment and control have equal mortality probabilities; each alternative says they differ.

2. Compare each ordered statistic with its own cutoff

$$\chi_j^2 > \chi_{\alpha_j,1}^2 \Rightarrow \text{reject } H_{0,j}$$
$$18.234103 > \chi_{.01666667,1}^2 = 5.731139$$
$$9.494427 > \chi_{.025,1}^2 = 5.023886$$
$$4.689514 > \chi_{.05,1}^2 = 3.841459$$

All three treatment–control differences are supported. Stop at the first comparison that fails; retain it and every remaining null. For t comparisons, sort $|t|$ largest first and use two-tail cutoff $t_{\alpha_j/2,\nu}$.

Read the supplied tables and test tails

t: row = df; select total two-tail α for a two-sided test. The equivalent one-tail column is $\alpha/2$. Compare $|t|$ with the positive cutoff; sign gives direction. **F:** numerator df across top, denominator df down side; right-tail α. **Chi-square:** row = df; right-tail α. The two supplied chi-square tables represent the same distribution; match the printed tail heading.

Missing column: obtain a more precise cutoff with the appropriate inverse distribution if available; use tail probability once. For example $t_{.00833333,42} = 2.493667$ for the Bonferroni two-sided $\alpha = .01666667$. A p-value smaller than alpha and a statistic beyond its matching critical value give the same decision.

Two proportions: thrombi and Yates z

Question. Thrombi occur in 18 of 25 placebo patients and 6 of 19 aspirin patients. Is the population thrombus probability different between groups at .05?

1. State the population hypotheses and sample fractions

$$H_0 : \pi_1 = \pi_2, \quad H_A : \pi_1 \neq \pi_2$$
$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{18}{25} = .72$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{6}{19} = .315789474$$

In words: placebo and aspirin have equal thrombus probabilities under the null; the probabilities differ under the alternative. x_i is the event count; n_i is that group's total.

2. Pool event counts for the null probability

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$
$$\hat{p} = \frac{18 + 6}{25 + 19} = \frac{24}{44} = .545454545$$

3. Calculate the null SE and half-count adjustment

$$SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$
$$SE_0 = \sqrt{.545454545(.454545455)\left(\frac{1}{25} + \frac{1}{19}\right)}$$
$$= .151547046$$
$$c_Y = \frac{1}{2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$
$$c_Y = \frac{1}{2}\left(\frac{1}{25} + \frac{1}{19}\right) = .0463157895$$

4. Shrink the absolute observed gap, then standardize

$$|z_Y| = \frac{\max(0, |\hat{p}_1 - \hat{p}_2| - c_Y)}{SE_0}$$
$$D_Y = \max(0, |.72 - .315789474| - .0463157895)$$
$$= .357894737$$
$$|z_Y| = \frac{.357894737}{.151547046} = 2.36160813$$

5. Make the two-sided decision

$$|z_Y| > z_{\alpha/2} \Rightarrow \text{reject } H_0$$
$$2.36160813 > 1.959964$$

Table: Normal cumulative .975, or t-table infinite-df row total two-tail .05. Reject equal population thrombus probabilities; aspirin has the lower observed proportion.

Folate: every expected count and corrected cell

Question. Is malformation occurrence associated with folate use? Test at .05. Apply Yates to this 2×2 table.

Group	Event	No event	Total
Folate	4	471	475
No folate	27	498	525
Total	31	969	1000

1. State independence in the population

$$H_0 : \text{group and outcome are independent}$$
$$H_A : \text{group and outcome are associated}$$

In words: the malformation probability is the same with and without folate under the null; it differs under the alternative.

2. Build ALL expected counts from margins

$$E_{ij} = \frac{R_i C_j}{N}$$
$$E_{11} = \frac{475(31)}{1000} = 14.725$$
$$E_{12} = \frac{475(969)}{1000} = 460.275$$
$$E_{21} = \frac{525(31)}{1000} = 16.275$$
$$E_{22} = \frac{525(969)}{1000} = 508.725$$

R_i (row total); C_j (column total); N (grand total); O_{ij} (observed count).

3. Calculate EVERY corrected cell contribution

$$c_{ij} = \frac{\max(0, |O_{ij} - E_{ij}| - .5)^2}{E_{ij}}$$
$$c_{11} = \frac{(|4 - 14.725| - .5)^2}{14.725} = 7.10021222$$
$$c_{12} = \frac{(|471 - 460.275| - .5)^2}{460.275} = .227148172$$
$$c_{21} = \frac{(|27 - 16.275| - .5)^2}{16.275} = 6.42400154$$
$$c_{22} = \frac{(|498 - 508.725| - .5)^2}{508.725} = .205515013$$

4. Add contributions; compare on the right tail

$$\chi^2_Y = \sum_i \sum_j c_{ij}, \quad \nu = (r-1)(c-1)$$
$$\chi^2_Y = 7.10021222 + .227148172$$
$$+6.42400154 + .205515013 = 13.9568769$$
$$\nu = (2-1)(2-1) = 1$$
$$13.9568769 > \chi^2_{.05,1} = 3.841459$$

Reject independence: folate use and malformation occurrence are associated. $p = .00018705$. Expected counts use the null margins; the observed counts remain unchanged. Yates shrinks the absolute gap before squaring.

Three groups: uncorrected larger-table chi-square

Question. Compare depression occurrence across three groups. Observed event/no-event counts are (36,131), (41,132), (75,104). Test at .05.

1. State the hypotheses; total rows and columns

$$H_0 : \pi_1 = \pi_2 = \pi_3, \quad H_A : \text{not all } \pi_i \text{ equal}$$
$$(R_1, R_2, R_3) = (167, 173, 179)$$
$$(C_1, C_2) = (152, 367), \qquad N = 519$$

In words: all three population event probabilities are equal under the null; at least one differs under the alternative.

2. Calculate ALL six null expected counts

$$E_{ij} = \frac{R_i C_j}{N}$$
$$E_{11} = \frac{167(152)}{519} = 48.9094412$$
$$E_{12} = \frac{167(367)}{519} = 118.090559$$
$$E_{21} = \frac{173(152)}{519} = 50.6666667$$
$$E_{22} = \frac{173(367)}{519} = 122.333333$$
$$E_{31} = \frac{179(152)}{519} = 52.4238921$$
$$E_{32} = \frac{179(367)}{519} = 126.576108$$

3. Calculate ALL six contributions (no Yates)

$$c_{ij} = \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$c_{11} = \frac{(36 - 48.9094412)^2}{48.9094412} = 3.40739270$$
$$c_{12} = \frac{(131 - 118.090559)^2}{118.090559} = 1.41123621$$
$$c_{21} = \frac{(41 - 50.6666667)^2}{50.6666667} = 1.84429825$$
$$c_{22} = \frac{(132 - 122.333333)^2}{122.333333} = .763851045$$
$$c_{31} = \frac{(75 - 52.4238921)^2}{52.4238921} = 9.72229698$$
$$c_{32} = \frac{(104 - 126.576108)^2}{126.576108} = 4.02667341$$

4. Add and select the larger-table df

$$\chi^2 = \sum c_{ij}, \quad \nu = (r-1)(c-1)$$
$$\chi^2 = 3.40739270 + 1.41123621 + 1.84429825$$
$$+.763851045 + 9.72229698 + 4.02667341$$
$$= 21.1757486$$
$$\nu = (3-1)(2-1) = 2$$
$$21.1757486 > \chi^2_{.05,2} = 5.991465$$

Reject equal population event probabilities. The statistic alone does not identify a particular differing group. Yates is applied to 2×2 tables, not this 3×2 table.

One list: goodness of fit to specified probabilities

Question. Counts are 255, 125 and 90. Do they fit target proportions .60, .25 and .15 at .05?

1. State the target distribution; find expected counts

$$H_0 : (\pi_1, \pi_2, \pi_3) = (.60, .25, .15)$$
$$H_A : \text{at least one population proportion differs}$$
$$N = 255 + 125 + 90 = 470, \quad E_i = Np_i$$
$$E_1 = 470(.60) = 282$$
$$E_2 = 470(.25) = 117.5, \quad E_3 = 470(.15) = 70.5$$

In words: the population follows the specified target proportions under the null; it does not follow that distribution under the alternative.

2. Work EVERY category, then sum

$$c_i = \frac{(O_i - E_i)^2}{E_i}, \quad \chi^2 = \sum_i c_i$$
$$c_1 = \frac{(255 - 282)^2}{282} = 2.58510638$$
$$c_2 = \frac{(125 - 117.5)^2}{117.5} = .478723404$$
$$c_3 = \frac{(90 - 70.5)^2}{70.5} = 5.39361702$$
$$\chi^2 = 2.58510638 + .478723404 + 5.39361702$$
$$= 8.45744681$$

3. Use category count minus one

$$\nu = k - 1, \quad \chi^2 > \chi^2_{\alpha,\nu}$$
$$\nu = 3 - 1 = 2, \qquad 8.45744681 > 5.991465$$

Reject the target distribution at .05. No Yates; this is a one-list fit test. The target probabilities must sum to one. Here they are specified, not estimated from these counts.

Followed groups: risk, ARR and NNT

Question. In the aspirin study, event counts are 6/19 treated and 18/25 controls. Quantify the treatment effect.

1. Find BOTH event risks from group totals

$$r_g = \frac{\text{events in group } g}{\text{all people in group } g}$$
$$r_T = \frac{6}{19} = .315789474$$
$$r_C = \frac{18}{25} = .72$$

T (treatment); C (control). Risks compare events with all group members, including nonevents.

2. Divide for relative risk; subtract for reduction

$$RR = \frac{r_T}{r_C}, \qquad ARR = r_C - r_T$$
$$RR = \frac{6/19}{18/25} = .438596491$$
$$ARR = \frac{18}{25} - \frac{6}{19} = .404210526$$

Treatment risk is about .439 times control risk. The absolute reduction is .4042, or 40.42 percentage points; it is not the relative reduction.

3. Take the reciprocal of ABSOLUTE risk reduction

$$NNT = \frac{1}{ARR}$$
$$NNT = \frac{1}{.404210526} = 2.47395833 \Rightarrow 3$$

Round upward to a whole person: treat 3 people to prevent one additional event over the studied interval. Use proportions, not percentage numbers.

4. If treatment increases harm, use NNH

$$ARI = r_T - r_C, \quad NNH = \frac{1}{ARI}$$
$$ARI = .08 - .03 = .05, \qquad NNH = \frac{1}{.05} = 20$$

Folate: name the numerator risk

Compare no-folate risk with folate risk

$$r_g = \frac{x_g}{n_g}, \qquad RR_{A/B} = \frac{r_A}{r_B}$$
$$r_{\text{no folate}} = \frac{27}{525} = .0514285714$$
$$r_{\text{folate}} = \frac{4}{475} = .00842105263$$
$$RR_{\text{no/yes}} = \frac{27/525}{4/475} = 6.10714286$$
$$RR_{\text{yes/no}} = \frac{4/475}{27/525} = .163742690$$

No-folate risk is about 6.11 times folate risk; reversing the comparison gives the reciprocal. State which group is in the numerator.

Selected cases and controls: odds ratio

Question. Among cases, 50 were exposed and 14 unexposed. Among controls, 43 were exposed and 35 unexposed. Estimate the exposure odds ratio.

	Exposed	Unexposed
Cases	50	14
Controls	43	35

Divide exposed by unexposed in EACH selected group

$$OR = \frac{a/b}{c/d} = \frac{ad}{bc}$$
$$O_{\text{case}} = \frac{50}{14} = 3.57142857$$
$$O_{\text{control}} = \frac{43}{35} = 1.22857143$$
$$OR = \frac{50/14}{43/35} = \frac{50(35)}{14(43)}$$
$$= \frac{1750}{602} = 2.90697674$$

Cases have about 2.91 times the exposure odds of controls. Case–control selection fixes how many cases are sampled; these sample fractions do not estimate disease risk. Use OR for this design, not RR.

Design and supplied chi-square output

Prospective: follow exposure/treatment groups to outcomes → risks and RR. **Retrospective case–control:** select by outcome, then compare past exposure → OR. Randomization assigns treatments; blinding/masking hides assignments. Parallel groups receive different treatments; crossover has the same participants receive treatments in different periods. Sham mimics a procedure without its active component.

```
Pearson's Chi-squared test with
Yates' continuity correction
X-squared = 5.5772, df = 1
p-value = 0.0182
```

Read the reported statistic, df and p-value. $.0182 < .05$ rejects independence for the aspirin 2×2 table. The correction is already included. For the larger table: $\chi^2 = 21.176$, df 2, $p = .00002522$; for goodness of fit: $\chi^2 = 8.4574$, df 2, $p = .01457$. Interpret the population association or target proportions stated by that test.